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R Phi Gate

Video: R Phi Gate
4 min

Graded Assignment:
Assessment 7
Submitted

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Assessment 7

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1. What is the result of measuring a qubit in a superposition state $|+\rangle$ using a quantum simulator?

1 / 1 point

The qubit collapses to the state $|0\rangle$ with 100% probability.

The state of the qubit becomes completely random.

The qubit collapses to the state $|1\rangle$ with 100% probability.

The qubit remains in the state $|+\rangle$.

Correct

Correct: Measurement collapses a superposition state to one of the basis states with a probability defined by the superposition coefficients.

Due

Dec 18, 11:59 PM IST

Attempts

Unlimited

Retry

2. Which sequence of gates can effectively create an X gate using Hadamard (H) and CNOT (C) gates?

1 / 1 point

$H \rightarrow H \rightarrow C$

$H \rightarrow C \rightarrow H$

$H \rightarrow C \rightarrow C$

$C \rightarrow H \rightarrow C$

Correct

Correct: This sequence of gates can effectively create an X gate.

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3. Which of the following quantum gates can transform the state of a qubit?

1 / 1 point

Pauli-Z (Z) gate

Measurement (M) gate

Pauli-X (X) gate

Correct

Correct: The Pauli-X gate flips the state of a qubit.

Hadamard (H) gate

Correct

Correct: The Hadamard gate is used to create superposition states.

Identity (I) gate

4. Which quantum gate is used to create a superposition state in a quantum computer?

1 / 1 point

Pauli-X gate

Toffoli gate

CNOT gate

Hadamard gate

Correct

Correct: The Hadamard gate is used to create a superposition state in a quantum computer.

5. Select the correct statements about parameter gates in quantum computing.

1 / 1 point

They automatically correct qubit errors.

They can only be used for single qubit operations.

They help in defining quantum algorithms.

Correct

Correct: Parameter gates are essential in designing quantum algorithms.

They provide a way to perform variable rotations on qubits.

Correct

Correct: Parameter gates allow for variable rotations based on parameters.

They enable the creation of quantum gates with adjustable parameters.

Correct

Correct: Parameter gates allow the creation of gates with adjustable parameters.